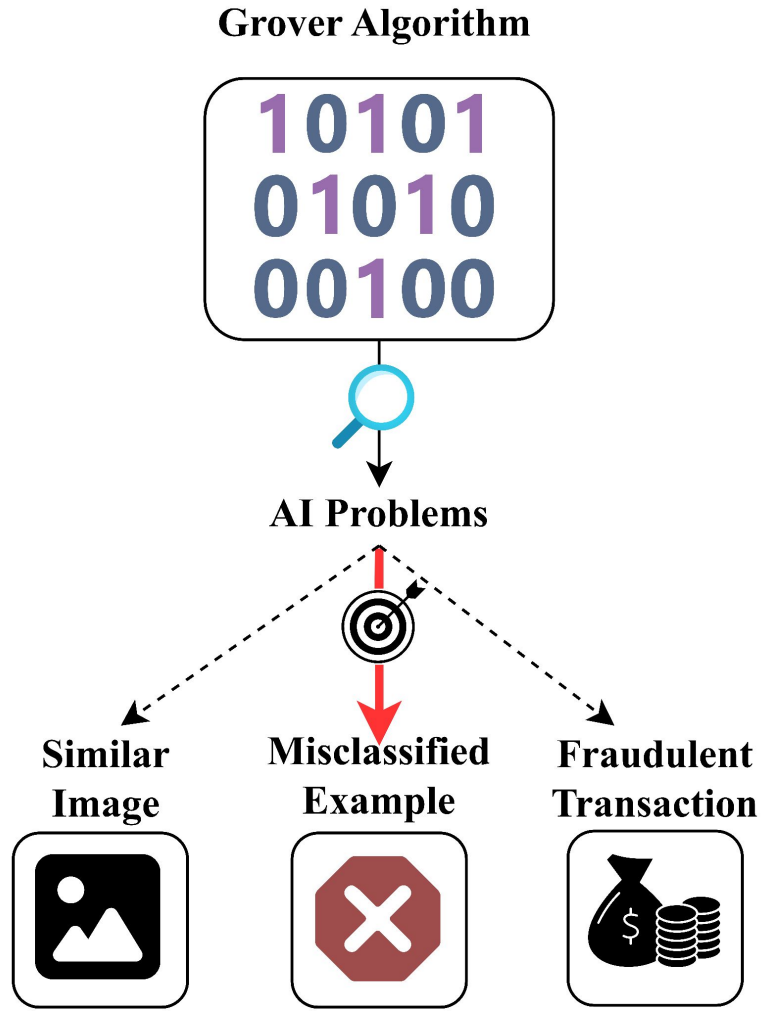
Grover's Algorithm for AI: A Quantum Approach to Unstructured Search

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What AI Problem Are We Tackling and Why Grover?

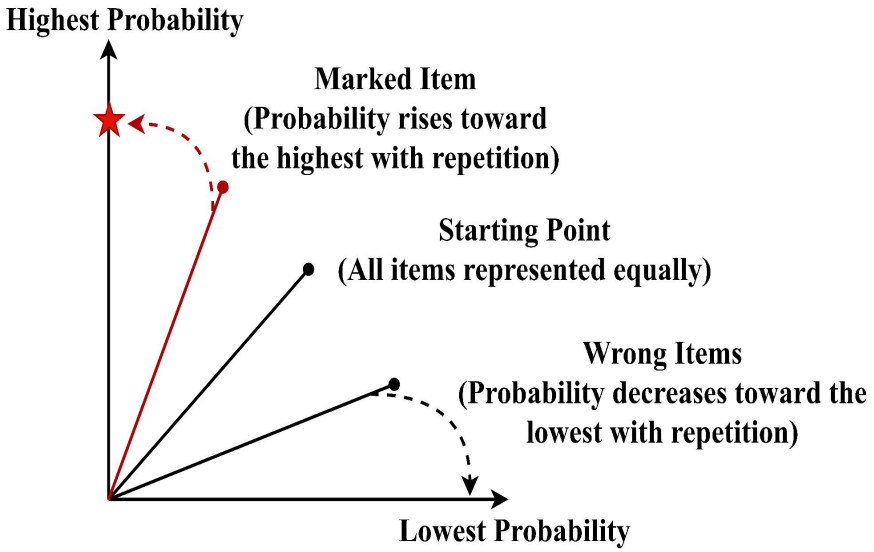
- Some AI tasks require searching through many examples when the desired example is not known in advance.
- For example:
  - Finding a misclassified example.
  - Finding a similar image.
  - Finding a fraudulent transaction.
- **Focus of this talk: Finding a misclassified example.**
- Grover's algorithm is designed for unstructured search.
- For N examples, finding a misclassified example using unstructured search requires:
  - **Classical search: O(N) queries.**
  - **Grover search: O(√N) queries.**

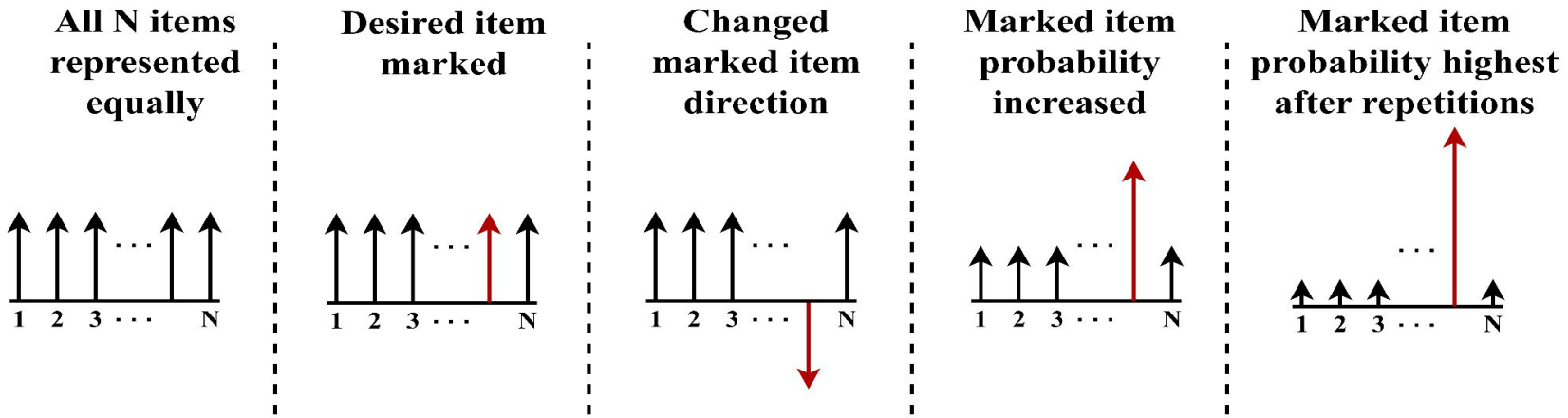


L. K. Grover, "A fast quantum mechanical algorithm for database search," in Proceedings of the twenty-eighth annual ACM symposium on Theory of computing, 1996, pp. 212–219.

How Grover's Algorithm Works

- Perform an unstructured search over N items.
- Apply a special quantum operation to represent all items equally.
- Use a quantum checking function to mark the desired item.
- Use another quantum operation to increase the probability of selecting the marked item.
- Repeat the marking and probability increasing steps $O(\sqrt{N})$ times.
- Measure to obtain the desired item with high probability.

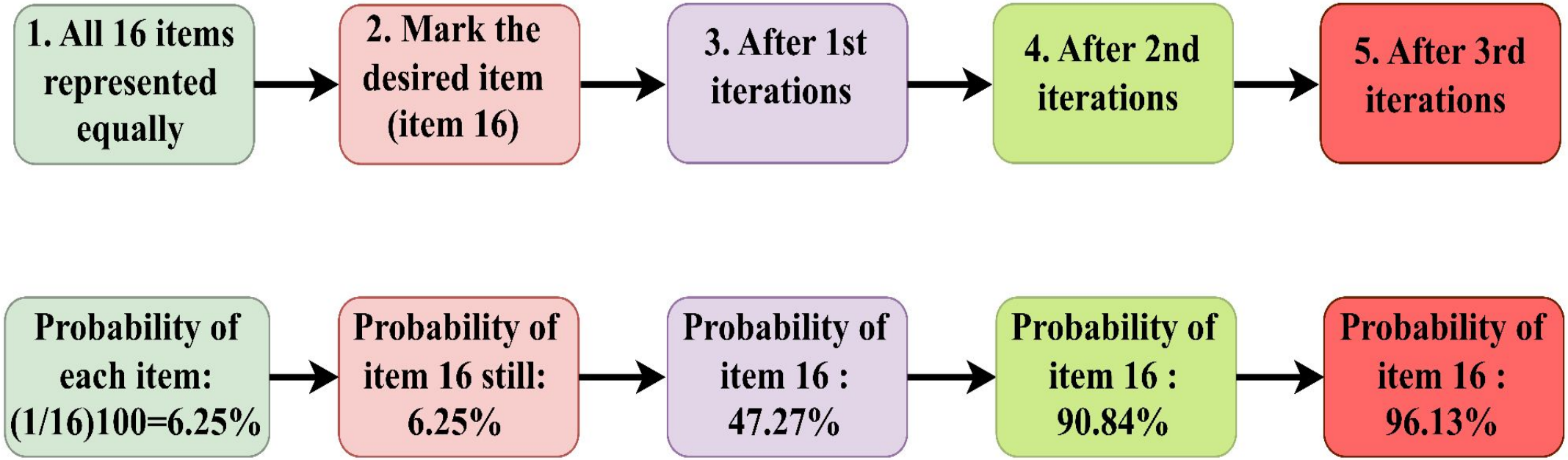




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Grover Search Example: N=16

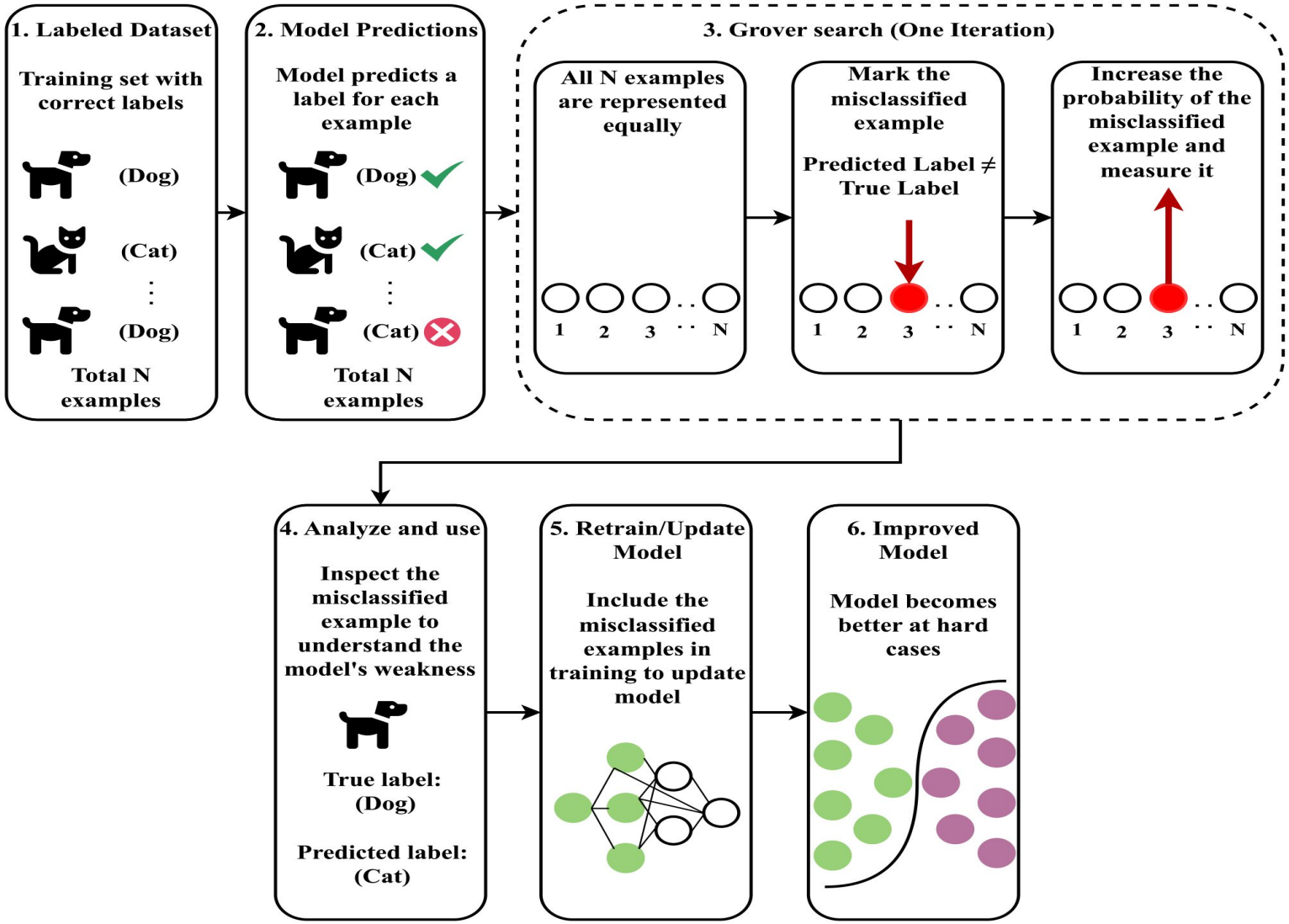
Suppose the desired item is: item 16



Grover Algorithm for AI: Finding Misclassified Examples

- Perform an unstructured search over N labeled examples.
- A classifier predicts a label for each example.
- Compare the predicted label with the known true label.
- A misclassified example satisfies: **Predicted Label ≠ True Label.**
- Use this condition to mark a misclassified example during Grover search.
- Measure to identify the misclassified example with high probability.
- Reuse the identified example during training to help improve the classifier.

Grover-Assisted AI: Working Flow Diagram



Limitations

- Requires a quantum computer to run Grover's algorithm.
- Requires an efficient quantum checking function for the search condition.
- $O(\sqrt{N})$ represents query complexity, not total runtime; actual runtime depends on the cost of the required quantum operations and may not always be faster than classical search.



Thank You

Any Questions?

Full Calculation of Grover's Algorithm for N=16

Grover's Algorithm : Full Calculation for $N = 16$

Marked item: item no. 16 $\Rightarrow$ $15 = 1111_2$ $\Rightarrow$ $\boxed{|w\rangle = |1111\rangle}$ $\qquad 2^4 = 16 \Rightarrow$ 4 qubits needed

Initial superposition: apply $H^{\otimes 4}$ to $|0000\rangle$: $\boxed{|\psi_0\rangle = \frac{1}{\sqrt{16}}\sum_{x=0}^{15}|x\rangle = \frac{1}{4}\sum_{x=0}^{15}|x\rangle}$ $\Rightarrow$ $P_0 = \left(\frac{1}{4}\right)^2 = \frac{1}{16} = \boxed{6.25\%}$

Each iteration: oracle flips the marked amplitude's sign, then diffusion reflects about the mean: $\boxed{a_i' = 2\bar{a} - a_i}$

Iteration 1

Oracle: $a_{\text{wrong}} = \frac{1}{4}, \quad a_{\text{marked}} = -\frac{1}{4}$

$$\bar{a}_1 = \frac{15\left(\frac{1}{4}\right) - \frac{1}{4}}{16} = \frac{7}{32}$$

$$a^{(1)}_{\text{wrong}} = 2\left(\frac{7}{32}\right) - \frac{1}{4} = \frac{3}{16}$$

$$a^{(1)}_{\text{marked}} = 2\left(\frac{7}{32}\right) + \frac{1}{4} = \frac{11}{16}$$

$$P_1 = \left(\frac{11}{16}\right)^2 = \frac{121}{256} =$$

$\boxed{47.27\%}$

Iteration 2

Oracle: $a_{\text{wrong}} = \frac{3}{16}, \quad a_{\text{marked}} = -\frac{11}{16}$

$$\bar{a}_2 = \frac{15\left(\frac{3}{16}\right) - \frac{11}{16}}{16} = \frac{17}{128}$$

$$a^{(2)}_{\text{wrong}} = 2\left(\frac{17}{128}\right) - \frac{3}{16} = \frac{5}{64}$$

$$a^{(2)}_{\text{marked}} = 2\left(\frac{17}{128}\right) + \frac{11}{16} = \frac{61}{64}$$

$$P_2 = \left(\frac{61}{64}\right)^2 = \frac{3721}{4096} =$$

$\boxed{90.84\%}$

Iteration 3

Oracle: $a_{\text{wrong}} = \frac{5}{64}, \quad a_{\text{marked}} = -\frac{61}{64}$

$$\bar{a}_3 = \frac{15\left(\frac{5}{64}\right) - \frac{61}{64}}{16} = \frac{7}{512}$$

$$a^{(3)}_{\text{wrong}} = 2\left(\frac{7}{512}\right) - \frac{5}{64} = -\frac{13}{256}$$

$$a^{(3)}_{\text{marked}} = 2\left(\frac{7}{512}\right) + \frac{61}{64} = \frac{251}{256}$$

$$P_3 = \left(\frac{251}{256}\right)^2 = \frac{63001}{65536} =$$

$\boxed{96.13\%}$

$$\boxed{P_0 = 6.25\% \longrightarrow P_1 = 47.27\% \longrightarrow P_2 = 90.84\% \longrightarrow P_3 = 96.13\%}$$

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